

Embedded Software Engineer with hands-on experience developing C firmware for STM32 and ESP32-S3 microcontrollers, including peripheral drivers, communication protocols, and finite-state machines. Experience in hardware-in-the-loop validation, hardware debugging, CI/CD, and robotics development. Master's graduate in Robotics Engineering from ENSTA Paris.

EXPERIENCE

ANSYS, Synopsys	may 2025 – oct 2025
Quality Assurance Engineering Intern	La Farlède, Provence-Alpes Côte d'Azur, France
Test Automation: Automated 43 manual regression tests into Azure Pipelines workflows using IronPython for the Speos optics simulation suite, covering LiDAR sensor and lens-modeling scenarios. The workflow built the target version, retrieved test inputs, executed simulations, compared outputs, and generated pass/fail reports, eliminating over 1,200 minutes of manual testing per release. Computer Vision: Developed from scratch a configurable Python image-comparison utility using OpenCV and scikit-image for regression testing, combining SSIM, Laplacian-based blur detection, color-histogram analysis, and template matching to tolerate small render variations.	
Skills : Azure Pipelines, CI/CD, Git, GitHub Actions, gRPC, IronPython, OpenCV, PyTest, Python, scikit-image	
SYSNAV	may 2024 – sep 2024
Embedded Firmware and Hardware Designer Intern	Vernon, Normandie, France
Low-Level Firmware Development: Developed an I2C firmware driver from scratch in C for a Texas Instruments battery-charger IC within an STM32H7 codebase, implementing register-level configuration, bitfield handling, status monitoring, and error handling for timeouts, NACKs, and invalid register states. Designed the driver's finite-state machine and validated over-current protection, thermal protection, and charge-state transitions against the device datasheet. Hardware-in-the-Loop Testing: Developed a Python hardware-in-the-loop test framework using an STM32H7 Nucleo board and 16 I2C-controlled relays to simulate connection faults between the station, base, and watch of the SYDE device. Designed the Python orchestration library and automated relay control and device-log retrieval. Reduced firmware validation time from 2 days to 2 hours by automating test setup and unattended overnight execution, covering 60% of relay combinations and reproducing field failures such as watch-detection faults.	
Skills : C, Git, GitLab, Hardware-in-the-Loop, I2C, Logic Analyzer, Oscilloscope, Python, STM32H7	
Amazon	sep 2023 – mar 2024
Retail Vendor Manager Intern	Clichy, Île-de-France, France
Data Processing Automation: Developed a VBA keyword-matching tool using regex, dictionaries, and array-based processing to classify brand-license data across 400,000+ entries, reducing processing time from 6 hours to 20 minutes.	
Skills : Business Intelligence, Data Analysis, Microsoft Excel, Process Optimization, SQL, VBA	
ISIR, Sorbonne Université	may 2023 – aug 2023
Neuroscience Research Intern	Paris, Île-de-France, France
Sensorimotor Simulation: Implemented a Python/Webots simulation environment based on mathematical models of hippocampal spatial cognition, including camera-based OpenCV analysis for navigation mapping; automated batches of 50+ simulation captures across 11 scenarios.	
Skills : Data Visualization, Git, LaTeX, Matplotlib, OpenCV, Python, Robotics Simulation, SciPy, Statistical Modeling, Webots	
Equipe Phoenix de Robótica da UNICAMP	jul 2020 – jul 2021
Head of Electronics	Campinas, São Paulo, Brazil
Leadership: Led a 12-member electronics division, developing Altium and electronics training for new members and coordinating board development, fabrication, and troubleshooting.	
Skills : Altium 365, Altium Designer, Hardware Debugging, PCB Design, Fabricaton and Assembly.	

PROJECTS

Equipe Paralela – Dolphin, an open-source sumo robot controller	feb 2026 – present
Embedded Software and Hardware Developer	Campinas, São Paulo, Brazil
Firmware: Developing C firmware for an ESP32-S3 robot controller using ESP-IDF, PlatformIO, CMake, and a minimal FreeRTOS layer, implementing table-driven finite-state-machine control for autonomous and radio-controlled operation. Validated the controller on a functional competition robot, finishing 6th of 11 robots in its latest competition. Hardware: Designed the main controller PCB, QRE1113 line sensor, and AS5147U encoder boards in KiCAD, integrating PWM motor control, ADC sensor acquisition, GPIO-controlled multiplexing, SPI peripherals, and interrupt-driven RC receivers and infrared sensors. Infrastructure: Built GitHub Actions CI for firmware compilation and KiCAD ERC/DRC validation.	
Skills : C, ESP32-S3, Git, GitHub Actions, KiCAD, PlatformIO.	

EDUCATION

École Nationale Supérieure de Techniques Avancées (ENSTA), Institut Polytechnique de Paris (IPP)	jul 2022 – oct 2025
Master's in Robotics Engineering	Palaiseau, Île-de-France, France
GPA: 3.71/4.00. Three-year double-degree program with UNICAMP.	
Faculdade de Engenharia Mecânica (FEM), Universidade Estadual de Campinas (UNICAMP)	feb 2019 – aug 2026
Bachelor's in Control and Automation Engineering	Campinas, São Paulo, Brazil
GPA: 0.84/1.00. Ranked 10th of 59 students and received the Academic Laurel for academic performance and on-time completion.	

LANGUAGES

English:	Fluent C1	French:	Fluent C1	Portuguese:	Native C2	Spanish:	Intermediate B1
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